

# Ritabrata Chakraborty

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## EDUCATION

### University of Pennsylvania

*M.S.E. in Robotics, GPA: -/4*

Pennsylvania, USA

*Aug '26 – May '28*

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

*B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10*

Rajasthan, India

*Oct '22 – May '26*

- **Institute Merit-cum-Need Scholarship** — Consecutive Recipient (Top 8% of Students)

## PUBLICATIONS & PEER REVIEWS

### Journal Publications

- **R. Chakraborty** and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- **R. Chakraborty** and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

### Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, **R. Chakraborty**, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- **R. Chakraborty**, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and **R. Chakraborty**, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

### Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

## EXPERIENCE – ROBOTICS & COMPUTER VISION

### Vision-Language Grounded Multi-Robot Coordination and Navigation

Supervisor: [Dr. Avinash Gautam](#), Associate Professor, BITS Pilani

*May '25 – Aug '25*

- Surveyed literature on vision-language grounded multi-robot coordination and VLMs-based spatial representations.
- Reviewed decentralized navigation strategies fusing 3D visual-language maps with LLM-driven instruction parsing.

### Multi-Critic Decomposition for Autonomous Indoor Navigation [🔗](#)

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist, CSIR-CEERI

*Jan '26 – May '26*

- Integrated three specialized critics for forward progress, lateral avoidance, and angular alignment with distinct reward signals.
- Deployed multi-critic TD3 with prioritized replay achieving 94.2% success rate and 22% lower collision rate versus baseline.

### Vision-Attention-Driven Autonomous Navigation with Semantic Understanding

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor, MARMoT Lab, NUS

*Aug '25 – Dec '25*

- Enhanced [CogniPlan](#) with cross-attention between frontier and node embeddings to enrich node representations.
- Incorporated [Visual Navigation Transformer \(ViNT\)](#) to capture semantic context for adaptive exploration strategies.

### Design and Development of Smart Automated Field Hockey Ball Launcher [🔗](#)

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor, BITS Pilani

*Apr '24 – Oct '25*

#### *Physics-Based Launcher Mechanical Design and FEM Analysis*

- Fabricated programmable launcher achieving 150 km/h using CFRP composite and stainless steel components.
- Performed FEM analysis and CAD using Fusion 360, achieving safety factor 2.07 at 496 RPM.

#### *Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification*

- Devised 3D ball localization from monocular videos using diameter-based depth and [PnLCalib](#) (0.76m RMSE).

- Constructed TCN-attention classifier fusing trajectory and statistical features, achieving 96.4% classification accuracy.

### Uncertainty-Guided Path Planning via Conditional Layout Prediction [↗](#)

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor, MARMoT Lab, NUS

Mar '25 – Aug '25

- Adapted multiple image-based exploration planners into ROS pipelines for evaluation in the [CMU Exploration Environment](#).
- Supported the development and benchmarking of [CogniPlan](#), achieving up to 17.7% shorter exploration paths and 3.9% improved navigation efficiency over state-of-the-art baselines across 100+ unseen maps.

### Monocular Vision-Based UAV Navigation for Orchard Monitoring [↗](#)

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist, CSIR-CEERI

Jan '24 – Feb '25

- Engineered a UAV-based orchard monitoring system using YOLOv11 (Box mAP50: 95.5%, Mask mAP50: 96.5%).
- Implemented B-spline trajectory logic and coded custom yaw-roll controller to minimize drift under wind.

## EXPERIENCE – GENERATIVE AI & PREDICTIVE MAINTENANCE

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### Synthetic Data Generation for Bearing Fault Diagnosis Under Data Scarcity

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor, KU Leuven

Sep '24 – May '26

#### Auxiliary Classifier WGAN-GP for Time-Series Sensor Data Generation [↗](#)

- Built ACWGAN-GP with TCN discriminator for minority fault class augmentation reaching ~100% classification accuracy.
- Assessed synthetic data using PCC, Cosine Similarity, MMD, and KL Divergence on FFT spectra.

#### Physics-Informed Generative Modeling for Bearing Fault Diagnosis Under Data Scarcity

- Formulated physics-informed cVAE-GAN and cVQ-GAN frameworks isolating latent-space effects in CWT scalogram synthesis.
- Validated [LiteFormer](#) classification gains across Density, Coverage, and CMMD metrics.

## EXPERIENCE – MECHANICAL SIMULATION & BEHAVIORAL RESEARCH

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### Finite Element Analysis of Running Track Surfaces

Supervisor: [Dr. Mani Shankar Dasgupta](#), Senior Professor, BITS Pilani

Apr '24 – Sep '25

- Modeled surface materials using hyperelastic formulations, evaluating restitution across 3 sports disciplines.
- Proposed optimized track designs improving durability by 23% based on ANSYS fatigue analysis.

### Primal World Beliefs in Engineering Students [↗](#)

Supervisor: [Dr. Rajneesh Choubisa](#), BITS Pilani

Aug '24 – Dec '24

- Conducted a primal world beliefs study surveying engineering undergraduates on core worldview dimensions.
- Replicated established population-level findings, validating trends against the original general-population study.

## INDUSTRY EXPERIENCE

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### ML Intern | Uber (AI Solutions)

Supervisor: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer, Uber

Jul '25 – Dec '25

- Streamlined object annotation and tracking in [uLabel](#) for RGB/IR video pipelines.
- Applied XGBoost anomaly detection (mAP50: 75%) to flag annotation errors during validation.

### Subject Matter Expert | Knowledge Cultivator

Self-supervised

Nov '23 – Feb '24

- Resolved numerical problems for student clients using MATLAB, CAD, and Python.
- Authored academic content across diverse research topics.

## PROJECTS

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### Vision-Guided Robotic Arm for Simulated Oral Cancer Screening

Aug '25 – Dec '25

- Simulated a KUKA KR 150 R3100 arm in MATLAB/Simulink for end-effector guidance inside a modeled oral cavity.
- Trained a vision model to detect cancerous regions, localizing the end effector through ORB-SLAM3 tracking.

### Autonomous Drone Navigation | MathWorks Global Student Drone Challenge 2025

Mar '25 – Apr '25

- Designed vision-based control for Parrot Mambo drone using masking, ray-tracing, and closed-loop yaw control.
- Optimized speed control and zone-based auto-landing to reduce track completion time.

### Air Quality Index Forecasting Using Time Series Models [↗](#)

Nov '24 – Nov '24

- Benchmarked AR, MA, ARMA, ARIMA, and SARIMA models forecasting daily and weekly AQI for Jaipur and Kolkata.
- Diagnosed stationarity with ADF and KPSS tests, tuning lag orders via ACF/PACF and scoring on  $R^2$ , MAE, and RMSE.

### 3D Lunar Surface Modeling from DEM Data | ISRO Immersion Challenge 2024 [↗](#)

Jul '24 – Jul '24

- Created lunar terrain models from TMC DEM data with curvature extraction, rendered live in UE5.
- Automated data import and heightmap transformation via a custom plugin, processing each tile in under 10s.

### Cricket Team Optimization | AMEX Campus Challenge 2024 (Product Track) [↗](#)

Jun '24 – Jul '24

- Processed and cleaned 100K+ datapoints to identify anomalies, outliers, and consistency trends.
- Generated optimal lineups using statistical heuristics, achieving a 14% gain over baseline strategies.

### ExoMy Rover Navigation and UR3 Arm Motion Planning | ERC 2023 Remote [↗](#)

Apr '23 – Sep '23

- Navigated ExoMy rover using ArUco detection, Ackermann steering, and spot turns for autonomous hazard avoidance.
- Calibrated UR3 arm with MoveIt and OMPL planner for collision-free manipulation (98% success).

## AWARDS & ACHIEVEMENTS

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| <b>Finalist</b> — Meta PyTorch OpenEnv Hackathon (India) <a href="#">🔗</a>                               | <i>Apr '26</i> |
| <b>3<sup>rd</sup> Place</b> — MathWorks Global Drone Student Challenge 2025                              | <i>Mar '25</i> |
| <b>Finalist</b> — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) <a href="#">🔗</a>      | <i>Jul '24</i> |
| <b>Top 15 Overall, Top 5 in College</b> — American Express Campus Challenge 2024 <a href="#">🔗</a>       | <i>Jul '24</i> |
| <b>5<sup>th</sup> Place &amp; Best Maintenance Award</b> — European Rover Challenge 2023 (Remote Finals) | <i>Sep '23</i> |

## TEACHING

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### Teaching Assistant

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| <i>CS F320: Foundations of Data Science, BITS Pilani</i>                                                                                                                                                                                                          | <i>Jan '26 – May '26</i> |
| <i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>                                                                                                                                                                                                         | <i>Jan '25 – May '25</i> |
| <i>ME F216: Materials Science and Engineering, BITS Pilani</i>                                                                                                                                                                                                    | <i>Sep '24 – Dec '24</i> |
| <ul style="list-style-type: none"><li>Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications.</li><li>Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.</li></ul> |                          |

## LEADERSHIP & VOLUNTEERING

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### Deputy Director of Academic Programming

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| <i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i> | <i>Aug '26 – Present</i> |
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### Vice-President

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| <i>South Asian Association (Rangoli), University of Pennsylvania</i> | <i>Aug '26 – Present</i> |
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### President & Secretary

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| <i>Mechanical Engineering Association (MEA), BITS Pilani</i>                                                                                                                                                                                                                        | <i>Jun '24 – May '26</i> |
| <ul style="list-style-type: none"><li>Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction.</li><li>Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.</li></ul> |                          |

### President & Tech Fest Coordinator

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| <i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i>                                                                                                                                                  | <i>Oct '24 – Jul '25</i> |
| <ul style="list-style-type: none"><li>Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts.</li><li>Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.</li></ul> |                          |

### Project Manager

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| <i>Tinkerer's Lab (TL), BITS Pilani</i>                                                                                                                                                                                                                          | <i>May '24 – Jul '25</i> |
| <ul style="list-style-type: none"><li>Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects.</li><li>Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.</li></ul> |                          |

### Treasurer

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| <i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i>                                                                                                 | <i>May '24 – Jul '25</i> |
| <ul style="list-style-type: none"><li>Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.</li></ul> |                          |

### Volunteer

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| <i>Junoon, National Service Scheme (NSS), BITS Pilani</i>                                                                                      | <i>Sep '23 – Sep '23</i> |
| <ul style="list-style-type: none"><li>Organized sports and recreational competitions for specially abled children from 25+ colleges.</li></ul> |                          |

## TECHNICAL SKILLS & COURSEWORK

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**Relevant Coursework:** Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins

**Programming Languages:** Python, C++, C, C#, Java, Bash (Shell)

**Robotics & Simulation:** ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D

**Data Science & Visualization:** NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel

**Cloud & MLOps:** AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions

**Hardware & Embedded Systems:** NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR

**CAD & Mechanical Simulation:** ANSYS Mechanical, SolidWorks, Fusion 360